

FinTrust Digital Bank - Week 4 Refined SQL Scripts & Business Findings

This document contains the refined production-ready SQL scripts and core business findings for Week 4 of the FinTrust Digital Bank project (AnalystLab Africa Experience Lab Internship Programme).

Section 1: Production-Ready Refined SQL Scripts

Query 1: RFM Scoring & Dynamic Customer Segmentation

-- Query 1: RFM Scoring & Dynamic Customer Segmentation

-- Calculates Recency, Frequency, and Monetary scores using NTILE window functions

-- and assigns strategic customer segments for targeted lifecycle retention.

WITH customer_rfm_base AS (

SELECT

c.customer_id,

c.customer_name,

MAX(t.transaction_date) AS last_transaction_date,

COUNT(t.transaction_id) AS total_transactions,

SUM(t.amount) AS total_spend

FROM customers c

LEFT JOIN transactions t

ON c.customer_id = t.customer_id

```
AND t.status = 'COMPLETED'
```

```
GROUP BY c.customer_id, c.customer_name
```

```
),
```

```
rfm_scores AS (
```

```
SELECT
```

```
customer_id,
```

```
customer_name,
```

```
last_transaction_date,
```

```
total_transactions,
```

```
total_spend,
```

```
-- Recency: Higher score for more recent activity (shorter gap from reference date)
```

```
NTILE(5) OVER (
```

```
ORDER BY last_transaction_date ASC NULLS LAST
```

```
) AS recency_score,
```

```
-- Frequency: Higher score for higher volume of transactions
```

```
NTILE(5) OVER (
```

```
ORDER BY total_transactions ASC NULLS LAST
```

```
) AS frequency_score,
```

```
-- Monetary: Higher score for higher total monetary value
```

```
NTILE(5) OVER (
```

```
ORDER BY total_spend ASC NULLS LAST
```

```
) AS monetary_score
```

```
FROM customer_rfm_base

)

SELECT

    customer_id,

    customer_name,

    last_transaction_date,

    total_transactions,

    COALESCE(total_spend, 0) AS total_spend,

    recency_score,

    frequency_score,

    monetary_score,

    CASE

        WHEN recency_score >= 4 AND frequency_score >= 4 AND monetary_score >= 4 THEN
        'Champions'

        WHEN recency_score >= 3 AND frequency_score >= 3 AND monetary_score >= 3 THEN
        'Loyal Customers'

        WHEN recency_score >= 3 AND frequency_score <= 2 THEN 'Promising / Recent Users'

        WHEN recency_score <= 2 AND frequency_score >= 4 THEN 'At Risk (High Volume)'

        WHEN recency_score <= 2 AND frequency_score <= 2 THEN 'Hibernating / Lost'

        ELSE 'Needs Attention'

    END AS customer_segment

FROM rfm_scores
```

ORDER BY monetary_score DESC, frequency_score DESC;

Data Output Messages Notifications							
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	customer_id [PK] character varying (50)	customer_segment character varying (50)	r_score integer	f_score integer	m_score integer	composite_rfm_score integer	rfm_segment text
1	FT-C00023	Student	4	4	4	12	Champions / High Val...
2	FT-C00560	Everyday	4	4	4	12	Champions / High Val...
3	FT-C00387	Student	4	4	4	12	Champions / High Val...
4	FT-C01480	Everyday	4	4	4	12	Champions / High Val...
5	FT-C00615	SME	4	4	4	12	Champions / High Val...
6	FT-C01285	SME	4	4	4	12	Champions / High Val...
7	FT-C00651	Everyday	4	4	4	12	Champions / High Val...
8	FT-C00377	Student	4	4	4	12	Champions / High Val...
9	FT-C01346	Everyday	4	4	4	12	Champions / High Val...
10	FT-C00497	Everyday	4	4	4	12	Champions / High Val...
11	FT-C00828	Premium	4	4	4	12	Champions / High Val...
12	FT-C00959	Everyday	4	4	4	12	Champions / High Val...
13	FT-C00155	Premium	4	4	4	12	Champions / High Val...
14	FT-C01365	Student	4	4	4	12	Champions / High Val...
15	FT-C00209	SME	4	4	4	12	Champions / High Val...
16	FT-C00845	Student	4	4	4	12	Champions / High Val...
17	FT-C00907	Student	4	4	4	12	Champions / High Val...
18	FT-C00662	Premium	4	4	4	12	Champions / High Val...
19	FT-C00412	Premium	4	4	4	12	Champions / High Val...
20	FT-C00036	Everyday	4	4	4	12	Champions / High Val...
21	FT-C01296	Everyday	4	4	4	12	Champions / High Val...
22	FT-C00018	SME	4	4	4	12	Champions / High Val...
23	FT-C00317	Student	4	4	4	12	Champions / High Val...

Query 2: Channel Performance, Failure Rates & Processing Volume

-- 2. Channel Performance, Failure Rates & Processing Volume
 -- Purpose: Evaluates technical channel reliability and throughput.

```
SELECT
  channel,
  COUNT(transaction_id) AS total_transactions,
  SUM(CASE WHEN transaction_status = 'Successful' THEN 1 ELSE 0 END) AS
successful_transactions,
  SUM(CASE WHEN transaction_status = 'Failed' THEN 1 ELSE 0 END) AS failed_transactions,
```

```

ROUND(100.0 * SUM(CASE WHEN transaction_status = 'Failed' THEN 1 ELSE 0 END) /
NULLIF(COUNT(transaction_id), 0), 2) AS failure_rate_pct,
ROUND(AVG(amount_ngn), 2) AS avg_transaction_amount,
ROUND(SUM(CASE WHEN transaction_status = 'Successful' THEN amount_ngn ELSE 0 END), 2)
AS total_successful_volume
FROM transactions
GROUP BY channel
ORDER BY total_successful_volume DESC;

```

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 5

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	channel character varying (50)	total_transactions bigint	successful_transactions bigint	failed_transactions bigint	failure_rate_pct numeric	avg_transaction_amount numeric	total_successful_volume numeric
1	Mobile App	5102	4579	296	5.80	47060.83	218678028.47
2	POS	2393	2174	125	5.22	43604.96	94671599.68
3	Web	1869	1698	88	4.71	47792.94	80504436.04
4	ATM	1747	1602	73	4.18	48049.43	77616712.15
5	USSD	889	803	48	5.40	48097.81	39336665.12

Query 3: Synthetic Risk Review Concentration by Segment & Channel

-- 3. Synthetic Risk Review Concentration by Segment & Channel
-- Purpose: Identifies operational risk hot spots across channels and demographics.

```

SELECT
    c.customer_segment,
    t.channel,
    COUNT(t.transaction_id) AS total_txns,
    SUM(CASE WHEN t.risk_review_flag = 'Yes' THEN 1 ELSE 0 END) AS flagged_txns,
    ROUND(100.0 * SUM(CASE WHEN t.risk_review_flag = 'Yes' THEN 1 ELSE 0 END) /
NULLIF(COUNT(t.transaction_id), 0), 2) AS risk_flag_pct,
    ROUND(SUM(CASE WHEN t.risk_review_flag = 'Yes' THEN t.amount_ngn ELSE 0 END), 2) AS
flagged_volume
FROM transactions t
JOIN customers c ON t.customer_id = c.customer_id
GROUP BY c.customer_segment, t.channel
ORDER BY risk_flag_pct DESC;

```

	customer_segment character varying (50) 🔒	channel character varying (50) 🔒	total_txns bigint 🔒	flagged_txns bigint 🔒	risk_flag_pct numeric 🔒	flagged_volume numeric 🔒
1	Premium	Web	361	92	25.48	7460280.02
2	Student	Web	352	80	22.73	6756178.37
3	Student	ATM	322	72	22.36	3455205.50
4	SME	ATM	226	50	22.12	3867516.56
5	Premium	ATM	353	77	21.81	6368148.96
6	Everyday	Web	899	184	20.47	12125677.47
7	Everyday	ATM	846	171	20.21	14501232.75
8	SME	Mobile App	707	140	19.80	10621635.17
9	Everyday	Mobile App	2368	461	19.47	33902570.31
10	Premium	Mobile App	996	192	19.28	9967605.71
11	SME	POS	376	72	19.15	4717841.85
12	Student	Mobile App	1031	197	19.11	13557083.58
13	Student	POS	419	78	18.62	4696928.94
14	SME	USSD	140	26	18.57	981952.79
15	Premium	POS	478	88	18.41	5042615.45
16	Everyday	POS	1120	201	17.95	12150215.33
17	Premium	USSD	173	31	17.92	1602261.98
18	Everyday	USSD	411	72	17.52	5415530.26
19	SME	Web	257	43	16.73	3317723.60
20	Student	USSD	165	25	15.15	1275426.81

Query 4: Month-over-Month (MoM) Growth & Risk Flag Trends

-- 4. Month-over-Month (MoM) Growth & Risk Flag Trends

-- Purpose: Tracks volume trajectories and flags systematic risk acceleration.

WITH Monthly_Summary AS (

SELECT

DATE_TRUNC('month', transaction_datetime) AS txn_month,

COUNT(transaction_id) AS monthly_txns,

SUM(amount_ngn) AS monthly_volume,

SUM(CASE WHEN risk_review_flag = 'Yes' THEN 1 ELSE 0 END) AS monthly_flagged_txns

FROM transactions

WHERE transaction_status = 'Successful'

GROUP BY DATE_TRUNC('month', transaction_datetime)

)

SELECT

txn_month,

```

monthly_txns,
ROUND(monthly_volume, 2) AS monthly_volume,
ROUND(LAG(monthly_volume, 1) OVER (ORDER BY txn_month), 2) AS prev_month_volume,
ROUND(100.0 * (monthly_volume - LAG(monthly_volume, 1) OVER (ORDER BY txn_month)) /
    NULLIF(LAG(monthly_volume, 1) OVER (ORDER BY txn_month), 0), 2) AS
mom_volume_growth_pct,
    ROUND(100.0 * monthly_flagged_txns / NULLIF(monthly_txns, 0), 2) AS
monthly_risk_flag_rate_pct
FROM Monthly_Summary
ORDER BY txn_month ASC;

```

	txn_month timestamp without time zone	monthly_txns bigint	monthly_volume numeric	prev_month_volume numeric	mom_volume_growth_pct numeric	monthly_risk_flag_rate_pct numeric
1	2026-01-01 00:00:00	3720	171270028.52	[null]	[null]	18.39
2	2026-02-01 00:00:00	3394	160166567.87	171270028.52	-6.48	19.56
3	2026-03-01 00:00:00	3742	179370845.07	160166567.87	11.99	19.80

Query 5: Domestic vs. International Transaction Risk & Failure Profiling

```

-- -----
-- 5. Domestic vs. International Transaction Risk & Failure Profiling
-- Purpose: Compares baseline operational performance across geographic boundaries.
-- -----

```

```

SELECT
    international_transaction,
    COUNT(t.transaction_id) AS total_transactions,
    ROUND(AVG(t.amount_ngn), 2) AS avg_amount,
    ROUND(SUM(t.amount_ngn), 2) AS total_amount,
    SUM(CASE WHEN t.risk_review_flag = 'Yes' THEN 1 ELSE 0 END) AS total_risk_flagged,
    ROUND(100.0 * SUM(CASE WHEN t.risk_review_flag = 'Yes' THEN 1 ELSE 0 END) /
NULLIF(COUNT(t.transaction_id), 0), 2) AS risk_flag_rate_pct,
    ROUND(100.0 * SUM(CASE WHEN t.transaction_status = 'Failed' THEN 1 ELSE 0 END) /
NULLIF(COUNT(t.transaction_id), 0), 2) AS failure_rate_pct
FROM transactions t
GROUP BY international_transaction;

```

	international_transaction character varying (10)	total_transactions bigint	avg_amount numeric	total_amount numeric	total_risk_flagged bigint	risk_flag_rate_pct numeric	failure_rate_pct numeric
1	No	11520	46916.80	540481570.98	2175	18.88	5.36
2	Yes	480	41657.88	19995783.87	177	36.88	2.50

-- 6. High-Value Transaction (HVT) Statistical Outlier Identification

-- Purpose: Flags transactions exceeding 3 standard deviations from aggregate mean.

WITH Stats AS (

SELECT

AVG(amount_ngn) AS avg_amt,

STDDEV(amount_ngn) AS stddev_amt

FROM transactions

)

SELECT

t.transaction_id,

t.customer_id,

c.customer_segment,

t.amount_ngn,

t.channel,

t.risk_review_flag,

ROUND((t.amount_ngn - s.avg_amt) / NULLIF(s.stddev_amt, 0), 2) AS z_score

FROM transactions t

CROSS JOIN Stats s

JOIN customers c ON t.customer_id = c.customer_id

WHERE t.amount_ngn > (s.avg_amt + (3 * s.stddev_amt))

ORDER BY t.amount_ngn DESC;

	transaction_id character varying (50) 🔒	customer_id character varying (50) 🔒	customer_segment character varying (50) 🔒	amount_ngn numeric (15,2) 🔒	channel character varying (50) 🔒	risk_review_flag character varying (10) 🔒	z_score numeric 🔒
1	FT-T009675	FT-C01330	Everyday	693454.45	USSD	Yes	7.52
2	FT-T009536	FT-C00590	Student	693006.62	Mobile App	No	7.51
3	FT-T005580	FT-C01048	Everyday	686958.44	POS	No	7.44
4	FT-T005494	FT-C00533	SME	675336.55	Web	No	7.31
5	FT-T011582	FT-C00031	Premium	672369.02	Mobile App	No	7.27
6	FT-T008166	FT-C00811	Everyday	656198.52	Web	Yes	7.08
7	FT-T004601	FT-C00994	Student	652685.59	Web	No	7.04
8	FT-T005697	FT-C01486	Premium	651566.54	Mobile App	Yes	7.03
9	FT-T006408	FT-C00772	Student	650580.09	Mobile App	No	7.02
10	FT-T008088	FT-C00422	Premium	643678.47	Mobile App	No	6.94
11	FT-T002109	FT-C00988	Student	643079.27	Web	No	6.93
12	FT-T005008	FT-C00874	Premium	640656.10	ATM	Yes	6.90
13	FT-T004394	FT-C00146	Everyday	639440.16	Web	No	6.89
14	FT-T000211	FT-C01138	SME	636603.07	Mobile App	No	6.86
15	FT-T007066	FT-C01433	Everyday	634438.35	POS	No	6.83
16	FT-T001068	FT-C01414	Student	633906.07	POS	No	6.83
17	FT-T004173	FT-C00726	Everyday	630973.90	ATM	Yes	6.79
18	FT-T005941	FT-C00230	Everyday	627812.60	POS	No	6.75
19	FT-T001839	FT-C01322	Student	623610.60	Mobile App	No	6.71

Query 7: Customer Transaction Frequency & Velocity Tiering

-- 7. Customer Transaction Frequency & Velocity Tiering
-- Purpose: Groups users by activity throughput to quantify high-frequency value.

```
WITH Customer_Velocity AS (  
  SELECT  
    customer_id,  
    COUNT(transaction_id) AS txn_count,  
    SUM(amount_ngn) AS total_spent  
  FROM transactions  
  GROUP BY customer_id  
)  
Tiered_Velocity AS (  
  SELECT  
    customer_id,  
    txn_count,  
    total_spent,  
    CASE  
      WHEN txn_count >= 15 THEN 'High Velocity (>=15 Txns)'  
      WHEN txn_count BETWEEN 8 AND 14 THEN 'Medium Velocity (8-14 Txns)'  
      ELSE 'Low Velocity (<8 Txns)'  
    END AS velocity_tier  
  FROM Customer_Velocity  
)  
SELECT  
  velocity_tier,  
  COUNT(customer_id) AS total_customers,  
  ROUND(AVG(total_spent), 2) AS avg_customer_spend,  
  ROUND(AVG(txn_count), 1) AS avg_txn_count_per_cust  
FROM Tiered_Velocity  
GROUP BY velocity_tier  
ORDER BY avg_customer_spend DESC;
```

	velocity_tier text	total_customers bigint	avg_customer_spend numeric	avg_txn_count_per_cust numeric
1	High Velocity (>=15 Txns)	29	770898.51	16.3
2	Medium Velocity (8-14 Tx...	778	464730.69	9.9
3	Low Velocity (<8 Txns)	693	254777.52	5.5

Query 8: Digital Engagement Score vs. Financial Activity & Risk Correlation

-- 8. Digital Engagement Score vs. Financial Activity & Risk Correlation

-- Purpose: Evaluates if app/digital engagement correlates with higher spend/risk.

```
WITH Digital_Engagement_Tiers AS (  
  SELECT  
    c.customer_id,  
    c.customer_segment,  
    c.digital_engagement_score,  
    CASE  
      WHEN c.digital_engagement_score >= 80 THEN 'High Engagement (80-100)'  
      WHEN c.digital_engagement_score BETWEEN 50 AND 79.9 THEN 'Moderate Engagement  
(50-79)'  
      ELSE 'Low Engagement (<50)'  
    END AS engagement_tier,  
    COUNT(t.transaction_id) AS total_txns,  
    SUM(CASE WHEN t.transaction_status = 'Successful' THEN t.amount_ngn ELSE 0 END) AS  
total_spend,  
    SUM(CASE WHEN t.risk_review_flag = 'Yes' THEN 1 ELSE 0 END) AS flagged_txns  
  FROM customers c  
  LEFT JOIN transactions t ON c.customer_id = t.customer_id  
  GROUP BY c.customer_id, c.customer_segment, c.digital_engagement_score  
)  
SELECT  
  engagement_tier,  
  COUNT(customer_id) AS total_customers,  
  ROUND(AVG(total_txns), 1) AS avg_txns_per_customer,  
  ROUND(AVG(total_spend), 2) AS avg_spend_per_customer,  
  ROUND(100.0 * SUM(flagged_txns) / NULLIF(SUM(total_txns), 0), 2) AS risk_flag_pct  
FROM Digital_Engagement_Tiers  
GROUP BY engagement_tier  
ORDER BY avg_spend_per_customer DESC;
```

	engagement_tier text	total_customers bigint	avg_txns_per_customer numeric	avg_spend_per_customer numeric	risk_flag_pct numeric
1	High Engagement (80-100)	398	8.2	344686.72	19.93
2	Moderate Engagement (50-...	865	8.0	344359.10	19.54
3	Low Engagement (<50)	237	7.8	319626.60	19.25

Section 2: Core Business Findings & Executive Insights Table

Query #	Query Subject	What It Reveals for FinTrust Leadership
Query 1	RFM Segmentation	Customer Value Concentration: Separates top-tier transactors (Champions) from inactive profiles (Hibernating). Identifies high-value accounts requiring VIP retention strategies and exposes dormant accounts where acquisition costs are unrecovered.
Query 2	Channel Performance	Infrastructure Vulnerabilities: Isolates processing capacity and failure rates by delivery channel (USSD vs. Mobile App vs. Web). Disproportions in channel failure rates point directly to gateway instabilities, third-party switch timeouts, or network routing issues.
Query 3	Risk Concentration	Targeted Fraud Risk Mapping: Cross-analyzes customer tier against payment channel to reveal where risk flags cluster. Allows compliance and risk teams to apply targeted MFA / Step-up authentication rules to high-risk channel-segment pairs without adding friction to low-risk users.

Query #	Query Subject	What It Reveals for FinTrust Leadership
Query 4	MoM Growth & Risk	Business Trajectory & Risk Volatility: Tracks month-over-month revenue velocity alongside compliance flag rates. Sustained volume growth accompanied by rising flag rates signals that scaling is introducing exposure to synthetic fraud or non-compliant account activity.
Query 5	Geographic Profiling	Cross-Border Exposure: Quantifies average ticket size, completion rates, and flag rates for international transactions compared to domestic baselines. Highlights potential foreign exchange bottlenecks or stricter anti-money laundering (AML) triggers on cross-border rails.
Query 6	Outlier Identification	Statistical Anomaly Detection: Applies Z-Score calculations ($Z > 3.0$) to flag transaction amounts that diverge significantly from aggregate distribution norms. Provides automated, objective rules for real-time transaction monitoring systems and high-value clearing reviews.
Query 7	Velocity Tiering	Power User Dynamics: Measures spend concentration

Query #	Query Subject	What It Reveals for FinTrust Leadership
		across usage tiers. Revealing what percentage of revenue is generated by high-velocity transactors (>= 15 txns) vs. low-frequency users guides decisions on tiered transaction fee structures and loyalty incentives.
Query 8	Digital Engagement	Platform Loyalty Impact: Evaluates whether active digital app usage directly translates to higher financial yield and account balances. Demonstrating a positive relationship between engagement score and spend validates product investments in UX, in-app features, and mobile usability.